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# Pokemon

Research Question:

- To what extent are Pokemon card rarity and age associated with their current market value?
    - I will explore whether cards from older sets have higher prices and whether its rarity explains its price difference compared to those in newer sets. There are some cards that have been found to be graded at pristine condition and have valued at over $1 million in both older and new sets. This question matters to collectors, buyers, sellers, and even people interested in how collectible markets assign value. The data can be either empirical or falsifiable. Yet, I expect rare cards to have higher market prices than common cards, but I don't expect age itself account for it alone. I would be surprised if age alone was a strongest predictor in market value instead of it combined with rarity.

Data Sources and method:

- My primary source will be the Pokemon TCG API: https://scrydex.com with its documentation at https://scrydex.com/docs/pokemon/api-reference. The endpoint will be https://api.scrydex.com/pokemon/v1/en/cards to get cards in English. Scrydex provides card IDs, names, rarity, release dates, language, and prices when requested. Access requires an API key. An alternate to this might be pokewallet.io: https://pokewallet.io/api-docs as it is free. This API is connected to TCGPlayer to gather real time prices and contains over 50k cards on file, plus sealed product prices. This can be helpful in coming up with new questions.
- I will use Python requests to get requests, parse JSON, and use pandas to flatten nested expansion and variant data into a table. This follows chapter 3 for evaluating access and platform restrictions. Chapter 4 for JSON structure and parsing, and Chapter 10 for request response workflows pagination. I will check response codes, handle error, and page through results in batches of at most 100.
- I will compare median prices by rarity and release period, make boxplots and scatterplots of age versus price. My target is approx.. 300-500 cards sampled across expansion packs from 1999 through todays date in 2026, including a mix of rarity categories. This will be a one-time price snapshot and not a claim on past market prices. I will compare either raw/ungraded and graded conditions where API permits and exclude those without price and age.

Ethics and feasibility:

- I will use Scrydex API rather than to scrape market pages. This requires a monthly membership, of $29. I will need to sign up to get an API key and grants me 5k credits. I can use this to access all TGCs and RAW prices only. With each request I can do up to 100 cards per request. Pokewallet.io is free, so I may use that one as a primary. I still need to sign up to get an API key, but the rate limits are 100 per hour and 1k per day. With the free plan I still get access to all 50k pokemon cards, rarity and prices. Since I will be exploring cards and not individual, I don't come across the concern of gathering personal information. But in the case that does happen, I will limit the content exploration to the cards themselves and not sellers information.

- Record:

In [1]:
```
import requests

url = "https://api.pokewallet.io/search"

headers = {"X-API-Key": "pk_live_3c74754205aac87847dbd07faa5089b82b556b4b5b258c12"}

params = {"q": "charizard", "limit": 1}

response = requests.get(url, headers=headers, params=params)

print(response.status_code)
print(response.json())
```

```
200
{'query': 'charizard', 'results': [{'id': 'pk_c5f39ad6d3d26b027c9ab95c280d871af3ed5f6c8d7654df36c5b264f691a25de62c2a01189979ddfa3a74', 'card_info': {'name': 'Charizard', 'clean_name': 'Charizard', 'set_name': 'Arceus', 'set_code': 'AR', 'set_id': '1391', 'card_number': '1/99', 'rarity': 'Holo Rare', 'card_type': 'Fire', 'product_type': 'card', 'hp': '140.0', 'stage': 'Stage 2', 'card_text': "Fire Formation — Each of Charizard's attacks does 10 more damage for each R Pokemon on your Bench to your opponent's Active Pokemon (before applying Weakness and Resistance).  (Poke-BODY)", 'attacks': ['[R] Fire Wing (30)', '[1RR] Burning Tail (80) Discard a R Energy attached to Charizard.'], 'weakness': 'W+30', 'resistance': 'F-20', 'retreat_cost': '3.0', 'variants': ['Holofoil', 'Reverse Holofoil']}, 'images': {'languages': ['en']}, 'tcgplayer': {'prices': [{'low_price': 88, 'mid_price': 125.88, 'high_price': 1999.7, 'updated_at': '2026-10-25T09:30:24.889108', 'market_price': 113.45, 'direct_low_price': None, 'sub_type_name': 'Holofoil'}, {'low_price': 297.98, 'mid_price': 298.99, 'high_price': 1283.93, 'updated_at': '2026-10-25T09:30:24.889108', 'market_price': 72.31, 'direct_low_price': None, 'sub_type_name': 'Reverse Holofoil'}], 'url': 'https://www.tcgplayer.com/product/84191'}, 'cardmarket': {'product_name': 'Charizard Lv.60', 'prices': [{'avg': 55.5, 'low': 7, 'avg1': 110, 'avg7': 54.25, 'avg30': 34.49, 'trend': 53.72, 'updated_at': '2026-10-25T09:03:49.621841', 'variant_type': 'holo'}, {'avg': 86.29, 'low': 4.3, 'avg1': 47, 'avg7': 72.05, 'avg30': 56.35, 'trend': 86.2, 'updated_at': '2026-10-25T09:03:49.621841', 'variant_type': 'normal'}], 'product_url': 'https://www.cardmarket.com/en/Pokemon/Products/Singles/Arceus/Charizard-Lv60-AR1'}}], 'pagination': {'page': 1, 'limit': 1, 'total': 549, 'total_pages': 549}, 'metadata': {'total_count': 549, 'tcg': 471, 'cardmarket': 398, 'tcg_only': 151, 'cardmarket_only': 78, 'both_sources': 320, 'truncated': True}}
```

I will use card id, card name, set id/name, rarity, variant, market price, and timestamp.

I did not use ChatGPT for this proposal but might use it for later to troubleshoot my code as I begin my research. I will then annotate the parts in which I did use it and how I used it.